

C.4 Trace of Hessian and Jacobian Norm

As reported in Table 1 previous works related with label noise, theoretically derived the connection between the trajectory of SGD and a trajectory which regularizes either the trace of the Hessian or the Jacobian norm. The Hessian of a loss function $\mathcal{L}$ can be decomposed as:

$$\nabla^2 \mathcal{L}(\mathbf{w}) = \sum_{i=1}^N \left[\underbrace{\nabla h(x_i; \mathbf{w}) [\nabla_h^2 l(h(x_i; \mathbf{w}))] \nabla h(x_i; \mathbf{w})^\top}_{G_i(\mathbf{w})} + \underbrace{\sum_{c=1}^K [\nabla_h l(h(x_i; \mathbf{w}))]_c \nabla^2 h(x_i; \mathbf{w})}_{E_i(\mathbf{w})} \right].$$

Many works demonstrated empirically that the G_i is the dominant part of the Hessian decomposition and $\nabla^2 L(\mathbf{w}) \sim \sum_i G_i$ (Sagun et al., 2017). The Jacobian (J) norm instead is defined as:

$$\|J(\mathbf{w})\|_F^2 = \frac{1}{N} \sum_{i=1}^N \text{Tr} (\nabla h_{\mathbf{w}}(x_i) \nabla h_{\mathbf{w}}(x_i)^\top), \quad (7)$$

in the case of square loss, $\nabla_h^2 l = I$ where I is the identity matrix. Hence, $\text{Tr} (\nabla^2 L(\mathbf{w})) \sim \|J\|_F^2$. The similarity is an exact equality at an interpolating solution since $\nabla_h l(h(x_i; \mathbf{w})) = 0$ and therefore not much ambiguity is left regarding which quantity to study. However, in the case of classification, this fact does not hold. In particular, the trace of $\nabla_h^2 l$ can significantly deviate from the identity matrix and varies depending on the value of the training loss. Consequently, although the two quantities seem closely related even when using the CE, we opt for analyzing $\|J\|_F$. This choice is motivated by its lack of explicit dependence on the training loss, enabling straightforward comparisons between different solutions. In the following, we report an empirical comparison of the two quantities along the training trajectory. In Fig. (14) we report a comparison between the trace of the Hessian and the Jacobian norm for both the fine-tuned iterates and the EMA. We can observe that for the fine-tuned iterates, both quantities display a decreasing trend; nevertheless, the ranking appears to be inverted to what is expected from previous theoretical works, i.e. larger LR should lead to a stronger regularization effect. If we compare the EMA instead, we can see that both quantities are still decreasing along the iterations but for the Hessian we don't observe any meaningful ranking for the final solutions whereas the Jacobian norm is lower for higher LR.

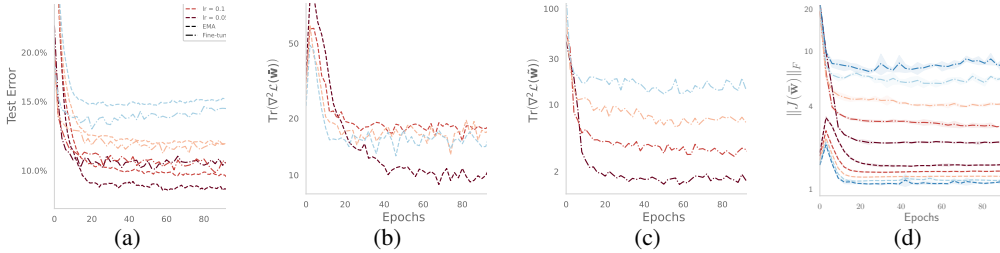


Figure 14: **Trace of Hessian and Jacobian Norm** We train standard Resnet18 on CIFAR-10 for 100 epochs fixing $\lambda = 0.0125$ and varying the learning rate. We report the EMA and we finetune for 100 epochs every 3 with $\eta = 10^{-3}$.

C.5 Experiments with scale-invariant Resnet on the sphere

In order to isolate the implicit regularization mechanism from the large initial drop and even small fluctuations in the dynamics of the ℓ_2 norm, we consider a simplified setting. We train scale-invariant networks (Li & Arora, 2019; Li et al., 2020) with projected SGD on the unitary sphere $\mathbb{S}^{(p-1)}$. We project the SGD iterates on the unitary sphere $\mathbb{S}^{(p-1)}$ within the context of scale-invariant ResNet architectures (Li & Arora, 2019; Li et al., 2020). This setup is helpful for two reasons: (a) it simplifies the selection of the LR and hence tremendously reduces the experimental overhead (b) scale-invariant networks have been extensively studied in previous works Li & Arora (2019); Li et al. (2020); Kodryan et al. (2022). Moreover, in the context of scale invariance, the optimization on the sphere is the natural object to study as the evolution of the direction is the only quantity that matters. The

projected SGD update writes as

$$\mathbf{w}_{t+1} = \Pi_{\mathbb{S}^{(p-1)}} (\mathbf{w}_t - \eta \nabla_{\mathbf{w}} \ell(y_{i_t}, h(\mathbf{w}_t, x_{i_t}))) \quad (8)$$

$$\text{where } \Pi_{\mathbb{S}^{(p-1)}} : \mathbf{w} \mapsto \mathbf{w} / \|\mathbf{w}\|_2. \quad (9)$$

The training framework still consists of two phases separated by a LR decay. The primary insight from our experiments on the sphere is depicted in Fig. 15: the test performance achieved in the fine-tuning phase depends on the LR used in the large-LR phase and, moreover, there is an optimal value. The work of Kodryan et al. (2022) reports a similar observation, where the best test loss is achieved at a LR where the loss neither converges too fast nor diverges but doesn't provide any explanation. Once again, our investigation reveals that the key to understand this behaviour and the dependence on the LR lies in the noisy dynamics in the large LR phase which closely tracks a regularized process. To summarize this idea we postulate a conjecture similar to the one reported in Section 2.2.

Conjecture 3. Consider the algorithm Eq. 8 with $\mathbf{w}_0$ initialized from a distribution $\mu_0(\mathbb{S}^{(p-1)})$. For any input x , let $\mathbf{w}_t, h(\mathbf{w}_t, x)$ be the random variables that denote the iterate at time t and its functional value. The stochastic process $(h(\mathbf{w}_t, x))_{t \in \mathbb{N}}$ will converge to a stationary distribution $\mu_\eta^\infty(x)$ with mean $\bar{\mu}_\eta(x)$ for which $\mathbf{w}_\eta^*$ is a first-order stationary point of the following regularized loss:

$$\bar{\mathcal{L}}(\mathbf{w}) := \mathcal{L}(\mathbf{w}) + \eta \sigma_\eta^2 \|J(\mathbf{w})\|_F^2. \quad (10)$$

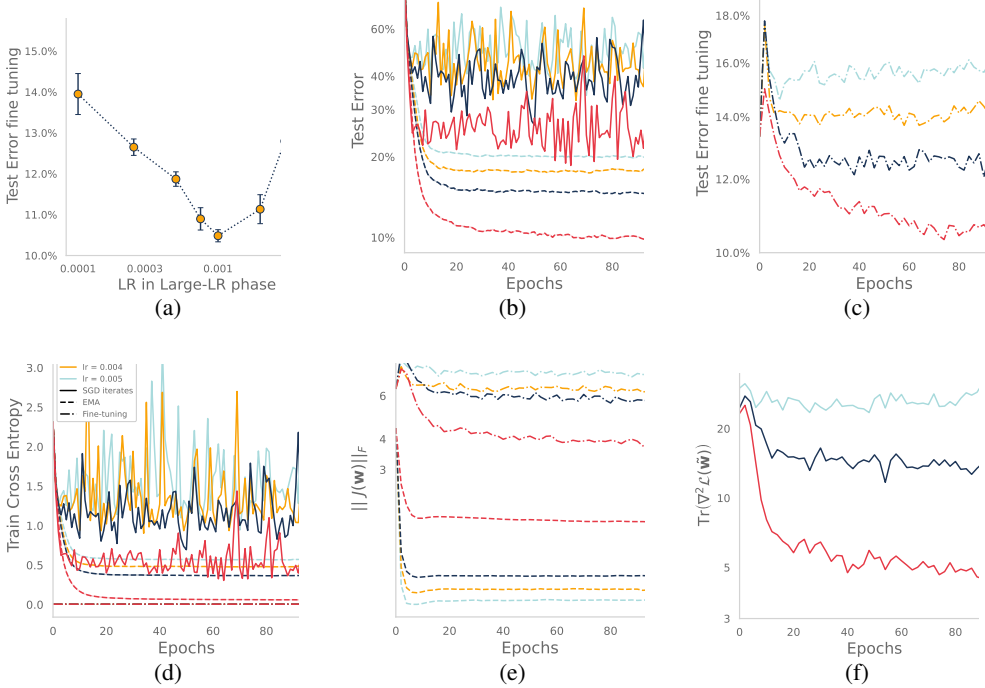


Figure 15: **Training scale-invariant ResNets on the sphere.** We train on CIFAR-10 with different large LR for the first 100 epochs and decay it to $\eta = 10^{-4}$ afterwards. Fig. (15a) reports the test error with respect to different LRs in the first phase showing the existence of an optimal value. Fig. (15b) reports the test error for the SGD iterates (—) and for the EMA (---). Figure (15c) reports the decreasing trend of the test error after fine-tuning for 100 epochs with $\eta = 10^{-4}$ every 2 epochs. Finally, Fig. (15e) reports the norm of the Jacobian for the EMA and the fine-tuned iterates and Figure (15f) reports a comparison with the trace of the Hessian for the fine-tuning iterates.

C.6 Empirical validation of Covariance approximation

We empirically verify the validity of the "decoupling approximation" introduced in Sec. 2.2 we use Stochastic Lanczos Quadrature (Yao et al., 2020) to estimate the empirical spectral density of the SGD covariance with and without the decoupling approximation, during the large-LR phase. The experiments are performed for ResNet-18 trained on the cifar10 dataset for different combinations of learning rate and weight decay which are used in the manuscript. The results in Figures 16, 17, and 18 illustrate a substantial overlap in the two spectra which serves as a validation of the reliability of our approximation.

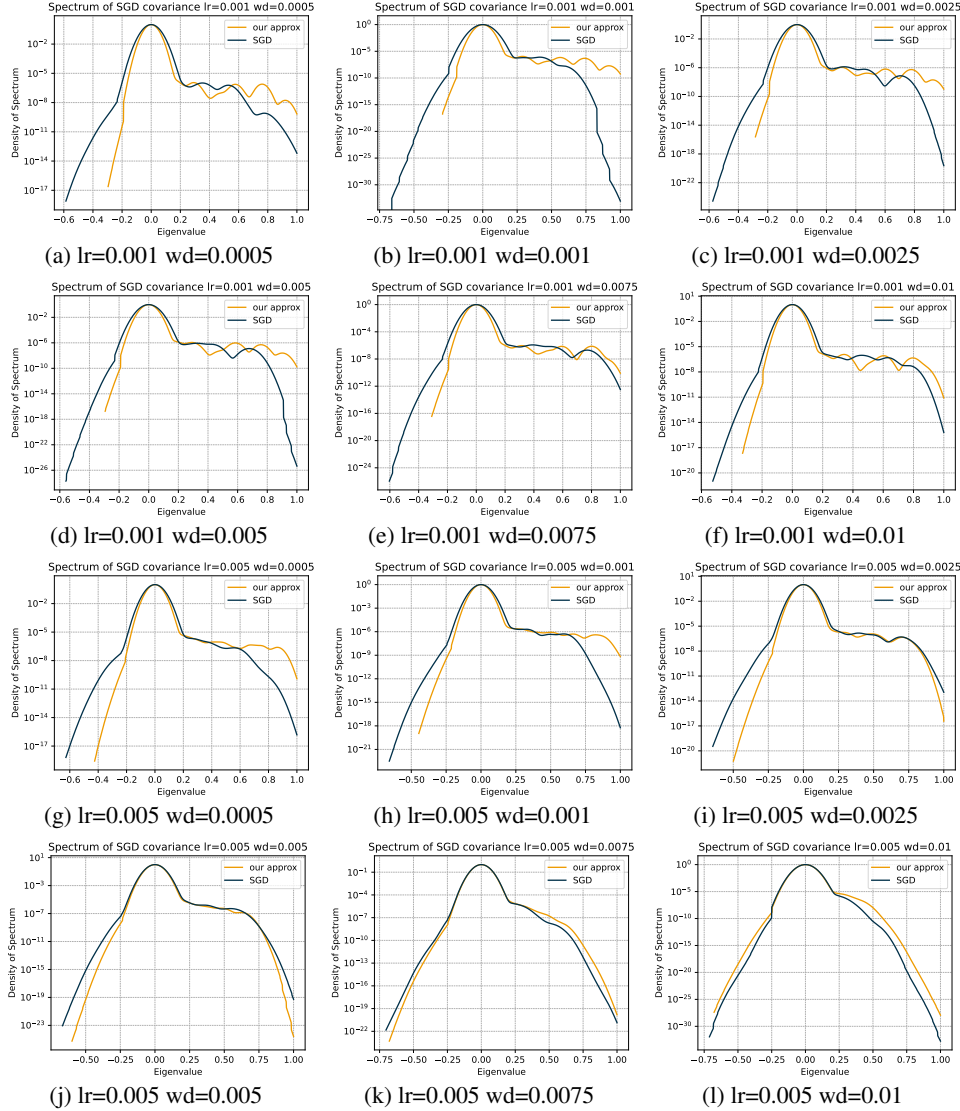


Figure 16: We use Stochastic Lanczos Quadrature to estimate the empirical spectral density of the SGD covariance, with and without the decoupling approximation, during the large-LR phase. Experiments with ResNet-18 on the CIFAR-10 dataset, varying learning rate and weight decay, show a substantial overlap in the two spectra, validating our approximation.